

AN EXPLICIT d^{-3} MOTION BOUND FOR PRIMITIVE DISTANCE-REGULAR GRAPHS

MACHINE-ASSISTED PROOF DEVELOPMENT

ABSTRACT. Let X be a primitive distance-regular graph on n vertices, of diameter $d \geq 3$. We prove, subject to the published inputs quoted precisely below, that either X is a Johnson graph or a Hamming graph, or

$$\text{motion}(X) \geq \frac{n}{12d^3}.$$

The published Pyber–Skresanov argument gives a bound on the scale n/d^6 in the primitive case. Four changes remove three powers of d and sharpen the constant: an exact formula for the vertices distinguishing an adjacent pair; use of the support-sensitive form of the geodesic boundary argument; retention of the full Metsch clique expression, which forces the geometric parameter m to satisfy $m \leq d$; and a direct geodesic Poincaré inequality that improves the spectral gap used in the $\mu = 1$ branch by a factor of eight. The remaining new step is an analytic sharpening of Kivva’s $\mu = 2$ multiplicity argument. It uses an exact recurrence for relative drops of the standard sequence and a direct concentration bound for the dominant distance sphere. Unlike earlier exploratory versions, the proof uses no finite tuple enumeration and no post-2021 classification theorem.

Verification status. The proof below is complete as written and no gap was found in a line-by-line source and algebra audit. Every external input is identified by theorem number (using the journal numbering), and all new scalar inequalities are checked by an accompanying exact-arithmetic script. This is nevertheless not independent peer review: the manuscript should not be cited as an established result until a specialist has checked the new $\mu = 2$ argument and the source interfaces.

1. STATEMENT AND PARAMETERS

Let X be a primitive distance-regular graph on n vertices, with valency k , diameter $d \geq 3$, intersection numbers b_i, c_i , and

$$\lambda = a_1, \quad \mu = c_2.$$

Write k_i for the size of a distance- i sphere, θ for the second largest adjacency eigenvalue, and $-m$ for the smallest eigenvalue, so $m > 0$.

Theorem 1.1. *Either X is a Johnson graph or a Hamming graph, or*

$$(1) \quad \boxed{\text{motion}(X) \geq \frac{n}{12d^3}.$$

Set

$$(2) \quad \gamma = \frac{1}{12d^3}, \quad \varepsilon = \frac{2}{12d^3 - 1}.$$

The parameters satisfy the exact closure identity

$$(3) \quad \frac{\varepsilon(1 - \gamma)}{2} = \gamma.$$

We shall repeatedly use

$$(4) \quad \gamma < \frac{1}{2}, \quad \varepsilon < \frac{2}{11d^3}, \quad d\varepsilon < \frac{1}{50}.$$

The last inequality is strongest at $d = 3$, where $d\varepsilon = 6/323 < 1/50$.

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2. ADJACENT DISTINGUISHERS AND AUTOMORPHISM SUPPORT

For vertices u, v , put

$$D(u, v) = \{z : \text{dist}(u, z) \neq \text{dist}(v, z)\}.$$

For adjacent u, v , the cardinality is constant; denote it by $D(1)$.

Lemma 2.1 (Exact adjacent-pair identity). *For every distance-regular graph of diameter at least three,*

$$(5) \quad D(1) = 2 + \frac{2}{k} \sum_{i=2}^d k_i c_i.$$

Consequently,

$$(6) \quad D(1) > \frac{\mu}{k} n.$$

Proof. For adjacent u, v , the number of vertices at distance i from both is $p_{i,i}^1$. The balance identity gives

$$k p_{i,i}^1 = k_i p_{1,i}^i = k_i a_i,$$

so

$$D(1) = n - \frac{1}{k} \sum_{i=1}^d k_i a_i.$$

Using $a_i = k - b_i - c_i$, $k_i b_i = k_{i+1} c_{i+1}$, $b_d = 0$, and $k_1 c_1 = k$, we obtain

$$\sum_{i=1}^d k_i a_i = k(n-2) - 2 \sum_{i=2}^d k_i c_i,$$

which proves (5).

We also use the standard comparison $c_i \leq b_j$ whenever $i + j \leq d$. Indeed, place vertices u, w, v in this order on a geodesic with $\text{dist}(u, w) = i$ and $\text{dist}(w, v) = j$. Every neighbor z of w with $\text{dist}(u, z) = i - 1$ must satisfy $\text{dist}(v, z) = j + 1$, so the c_i such neighbors form a subset of the b_j such neighbors. In particular $c_2 \leq b_1$, hence

$$k_2 = \frac{k b_1}{c_2} \geq k_1 = k.$$

It follows that $\sum_{i=2}^d k_i \geq (n-1)/2$. Since $c_i \geq c_2 = \mu$ for $i \geq 2$,

$$D(1) \geq 2 + \frac{\mu}{k} (n-1) > \frac{\mu}{k} n,$$

where the final inequality uses $\mu \leq k$. □

Lemma 2.2 (Support-sensitive adjacent pair). *Let $g \in \text{Aut}(X)$ be nonidentity, put $S = \text{supp}(g)$ and $\rho = |S|/n \leq 1/2$. Some $x \in S$ has at least*

$$(7) \quad \frac{k}{d} (1 - \rho)$$

fixed neighbors. If $\mu < k(1 - \rho)/d$, then $x \sim x^g$ and

$$(8) \quad \lambda \geq \frac{k}{d} (1 - \rho), \quad D(1) \leq |S|.$$

Proof. We spell out the orientation convention in the support-sensitive form of Pyber–Skresanov’s Proposition 2.8. For a directed edge e of X , let

$$P_e = \sum_{u,v} \frac{1}{p(u,v)} \# \{L : L \text{ is a } u\text{--}v \text{ geodesic using } e\},$$

where $p(u,v)$ is the number of u – v geodesics. Their coherent-configuration argument shows that P_e is independent of the directed edge; write the common value as P . Summing over the nk directed edges gives

$$nkP = \sum_{u,v} \text{dist}(u,v) \leq dn^2, \quad \text{so} \quad P \leq \frac{dn}{k}.$$

Let

$$\delta_X^+(S) = \{(u,v) : u \in S, v \notin S, u \sim v\}.$$

For each ordered pair $(x,y) \in S \times (V(X) \setminus S)$, every x – y geodesic uses at least one edge of $\delta_X^+(S)$. Consequently

$$|S|(n - |S|) \leq |\delta_X^+(S)|P,$$

and hence

$$(9) \quad |\delta_X^+(S)| \geq |S| \frac{k}{d} \frac{n - |S|}{n}.$$

Each undirected cut edge contributes exactly one directed edge to $\delta_X^+(S)$, so this is also the corresponding undirected cut size. Averaging the outbound cut degrees over the vertices of S proves (7). Since S is the support of g , every neighbor outside S is fixed.

Every fixed neighbor of x is also adjacent to x^g . If x and x^g were nonadjacent, the existence of a common neighbor would put them at distance two, where they have exactly μ common neighbors. This contradicts the strict hypothesis. Thus $x \sim x^g$, and the fixed common neighbors give the lower bound on λ .

For $z \notin S$,

$$\text{dist}(x, z) = \text{dist}(x^g, z^g) = \text{dist}(x^g, z),$$

so no fixed vertex distinguishes x and x^g . Hence $D(1) \leq |S|$. $\square$

3. A DIRECT GEODESIC POINCARÉ INEQUALITY

The next proposition is the self-contained spectral improvement used in the $\mu = 1$ branch.

Proposition 3.1 (Geodesic spectral-gap bound). *Let G be the graph of a connected symmetric basis relation of valency k in a homogeneous coherent configuration on n points. If D is its diameter and θ_1 its second largest adjacency eigenvalue, then*

$$(10) \quad k - \theta_1 \geq \frac{n^2 k}{\sum_{x,y \in V(G)} \text{dist}(x,y)^2} \geq \frac{k}{D^2}.$$

For a distance-regular graph,

$$(11) \quad k - \theta_1 \geq \frac{nk}{\sum_{i=0}^D k_i i^2}.$$

Proof. For ordered vertices x, y , let $p(x, y)$ be the number of geodesics from x to y . For a directed edge e , define

$$Q_e = \sum_{x,y} \frac{\text{dist}(x,y)}{p(x,y)} \# \{P : P \text{ is an } x\text{--}y \text{ geodesic containing } e\}.$$

The intersection-number argument in Pyber–Skresanov’s proof of Proposition 2.8 shows that Q_e is independent of e . Here is the precise interface: after fixing the three basis relations containing (x, z) , (w, y) , and (x, y) for $e = (z, w)$, the number of relevant pairs and geodesics is independent of (z, w) by their Lemma 2.7; the extra weight $\text{dist}(x, y)/p(x, y)$ depends only on the basis relation containing (x, y) . Write the common value as Q .

Summing over the nk directed edges gives

$$(12) \quad nkQ = \sum_{x,y} \text{dist}(x, y)^2,$$

because every geodesic of length ℓ contributes ℓ at each of its ℓ directed edges.

Let f have mean zero. Averaging Cauchy–Schwarz along all geodesics from x to y gives

$$(f(x) - f(y))^2 \leq \frac{\text{dist}(x, y)}{p(x, y)} \sum_{P: x \rightarrow y} \sum_{e \in P} (\nabla_e f)^2.$$

After summing over ordered pairs and using the constancy of Q_e ,

$$2n\|f\|_2^2 \leq Q \sum_{e \text{ directed}} (\nabla_e f)^2 = 2Q f^\top (kI - A)f.$$

Taking f in the θ_1 -eigenspace gives $k - \theta_1 \geq n/Q$. Equation (12) proves the first inequality in (10); the bound $\text{dist}(x, y) \leq D$ proves the second. In a distance-regular graph,

$$\sum_{x,y} \text{dist}(x, y)^2 = n \sum_{i=0}^D k_i i^2,$$

which proves (11). □

Remark 3.2. Pyber and Skresanov obtain $k - \theta_1 \geq k/(8D^2)$ by passing from their geodesic expansion estimate through Cheeger’s inequality. Proposition 3.1 applies Cauchy–Schwarz directly to the same uniform geodesic loads and retains the squared path lengths. No novelty claim is made for this canonical-path inequality without a broader literature search.

4. SMALL SUPPORT FORCES DELSARTE GEOMETRY

The proof of Pyber–Skresanov’s Proposition 2.6 retains the following consequence of Metsch’s theorem. If $\lambda^2 \geq 4k\mu$, then the graph contains a clique of size at least

$$(13) \quad \lambda + 2 - \left(\left\lceil \frac{3k}{2(\lambda + 1)} \right\rceil - 1 \right) (\mu - 1).$$

Proposition 4.1 (Structural reduction). *Suppose that a nonidentity automorphism has support density $\rho < \gamma$. Then*

$$(14) \quad \mu < \gamma k, \quad \lambda > \frac{1 - \gamma}{d} k, \quad k > 12d^3,$$

and X is Delsarte-geometric with smallest eigenvalue $-m$ satisfying

$$(15) \quad m \leq d.$$

Proof. Suppose first that $\mu \geq k/(2d)$. Let $k_i = k_{\max}$ be a largest nontrivial relation valency. Since $k_2 \geq k_1$, we may take $i \geq 2$. Then

$$\frac{k_i}{k_{i-1}} = \frac{b_{i-1}}{c_i} \leq \frac{k}{\mu} \leq 2d,$$

so $n - k_{\max} \geq k_{i-1} \geq k_{\max}/(2d)$. Every nontrivial relation of the primitive distance scheme has diameter at most d ; hence Pyber–Skresanov Propositions 2.10 and 2.12 give

$$\text{motion}(X) \geq \frac{n - k_{\max}}{d}.$$

If $k_{\max} \geq n/2$, this is at least $n/(4d^2)$; otherwise it is greater than $n/(2d)$. Both bounds exceed γn , contradicting $\rho < \gamma$. Therefore

$$(16) \quad \mu < \frac{k}{2d}.$$

Since $\rho < \gamma < 1/2$, Lemma 2.2 applies and gives

$$\lambda > \frac{1 - \gamma}{d} k.$$

Together with Lemma 2.1 and $D(1) \leq |S|$, it gives

$$\mu < \rho k < \gamma k.$$

As $\mu \geq 1$, this also implies $k > 1/\gamma = 12d^3$.

Put $\alpha = (1 - \gamma)/d$. We need three elementary inequalities:

$$(17) \quad \alpha^2 > 4\gamma, \quad \alpha - \frac{3\gamma}{2\alpha} > \frac{1}{d+1}, \quad (d+1)^2\gamma < \alpha.$$

For the first, $1 - \gamma > 1/2$ gives $\alpha^2 > 1/(4d^2) > 1/(3d^3) = 4\gamma$. For the second,

$$d^2 \left(\alpha - \frac{1}{d+1} - \frac{3\gamma}{2\alpha} \right) = \frac{d}{d+1} - \frac{1}{12d^2} - \frac{1}{8(1-\gamma)} > \frac{3}{4} - \frac{1}{108} - \frac{1}{4} > 0.$$

For the third,

$$d(d+1)^2\gamma = \frac{(d+1)^2}{12d^2} \leq \frac{4}{27} < \frac{1}{2} < 1 - \gamma.$$

The first inequality and (14) give $\lambda^2 > 4k\mu$, so (13) applies. Let L be the resulting clique size. Since $\lceil x \rceil - 1 < x$ for every real x ,

$$\begin{aligned} L - 1 &\geq \lambda + 1 - \left(\left\lceil \frac{3k}{2(\lambda+1)} \right\rceil - 1 \right) (\mu - 1) \\ &> \alpha k - \frac{3k}{2(\lambda+1)} \mu > k \left(\alpha - \frac{3\gamma}{2\alpha} \right) > \frac{k}{d+1}. \end{aligned}$$

The Delsarte clique bound $L \leq 1 + k/m$ therefore gives $m < d + 1$. Finally,

$$m^2\mu < (d+1)^2\gamma k < \alpha k < \lambda.$$

By the Bang–Koolen criterion, quoted as Pyber–Skresanov Proposition 2.5, X is Delsarte-geometric. In a Delsarte geometry m is the integer number of Delsarte cliques through each vertex, so $m \leq d$. $\square$

5. THE TRANSITION AND SPECTRAL BRANCHES

Assume from now on that a nonidentity automorphism has support density $\rho < \gamma$. Proposition 4.1 is in force.

Lemma 5.1 (Transition without a diameter loss). *If*

$$b_j \geq \varepsilon k, \quad c_{j+1} \geq \varepsilon k$$

for some $1 \leq j \leq d-1$, then $|S| > \varepsilon n > \gamma n$.

Proof. For $i \leq j$, monotonicity gives $b_i \geq \varepsilon k$; for $i \geq j + 1$, it gives $c_i \geq \varepsilon k$. Thus $a_i \leq (1 - \varepsilon)k$ for every $i \geq 1$, and

$$D(1) = n - \frac{1}{k} \sum_{i=1}^d k_i a_i \geq n - (1 - \varepsilon)(n - 1) > \varepsilon n.$$

Now use $D(1) \leq |S|$ from Lemma 2.2. $\square$

If no transition occurs, let t be the least index with $b_t \leq \varepsilon k$, using the convention $b_d = 0$. The inequality $2\lambda \leq k + \mu$ gives

$$b_1 = k - \lambda - 1 \geq \frac{k - \mu - 2}{2} > \frac{1 - 3\gamma}{2}k > \frac{k}{3},$$

where $1/k < \gamma$ was used. Hence $t \geq 2$. Since $b_{t-1} > \varepsilon k$ and no transition occurs,

$$(18) \quad b_t \leq \varepsilon k, \quad c_t < \varepsilon k.$$

Lemma 5.2 (High second eigenvalue). *Under these assumptions,*

$$(19) \quad \theta \geq (1 - \varepsilon)b_1.$$

Proof. Suppose instead that $\theta < (1 - \varepsilon)b_1$. Since $m \leq d$, $b_1 > k/3$, $k > 12d^3$, and $\varepsilon < 1/2$, we also have $m < (1 - \varepsilon)b_1$. Thus the zero-weight spectral radius $\xi = \max\{\theta, m\}$ satisfies $\xi < (1 - \varepsilon)b_1$.

Moreover $\lambda > \mu$, because $(1 - \gamma)/d > \gamma$. Every pair of vertices has at most λ common neighbors. Babai's spectral motion bound, Pyber–Skresanov Proposition 2.13, therefore gives

$$\begin{aligned} \rho &\geq \frac{k - \xi - \lambda}{k} > \frac{1 + \varepsilon b_1}{k} \\ &= \frac{1 - \varepsilon}{k} + \varepsilon \frac{k - \lambda}{k} \geq \frac{1 - \varepsilon}{k} + \frac{\varepsilon}{2} \left(1 - \frac{\mu}{k}\right) \\ &> \frac{\varepsilon}{2}(1 - \gamma) = \gamma, \end{aligned}$$

contradicting $\rho < \gamma$. The last equality is (3). $\square$

6. THE PUBLISHED JOHNSON AND $\mu = 1$ ENDGAMES

Proposition 6.1 (The case $\mu \geq 3$). *Under (18) and (19), if $\mu \geq 3$, then X is a Johnson graph.*

Proof. If the neighborhood graphs are disconnected, Pyber–Skresanov Proposition 2.20 gives

$$\theta + 1 \leq \frac{5}{7}b_1,$$

contradicting (19), since $\varepsilon < 2/7$. Hence the neighborhood graphs are connected.

Pyber–Skresanov Proposition 2.19 supplies an absolute constant $\varepsilon_* > 0.0065$ such that a Delsarte-geometric distance-regular graph with $\mu \geq 2$, a connected neighborhood graph,

$$\theta + 1 > (1 - \varepsilon_*)b_1, \quad k \geq \max\{m^3, 29\},$$

is Johnson. Here $\varepsilon \leq 2/323 < 13/2000 = 0.0065 < \varepsilon_*$ and (19) give the spectral condition, while $k > 12d^3 \geq 12m^3$ gives the valency condition. $\square$

Proposition 6.2 (The case $\mu = 1$). *Under Proposition 4.1, the case $\mu = 1$ contradicts $\rho < \gamma$.*

Proof. Proposition 3.1 gives

$$\theta \leq k \left(1 - \frac{1}{d^2}\right).$$

Since $m \leq d$ and $k > 12d^3$, the zero-weight spectral radius satisfies

$$\xi = \max\{\theta, m\} \leq k(1 - \eta), \quad \eta = \frac{1}{d^2} < \frac{1}{2}.$$

If $m \geq 3$, then

$$k > 12d^3 \geq 4md^2 = \frac{4m}{\eta}, \quad k > m^2.$$

Pyber–Skresanov Proposition 2.14 therefore yields

$$\text{motion}(X) \geq \frac{\eta n}{4} = \frac{n}{4d^2} > \gamma n.$$

If $m < 3$, integrality gives $m = 1$ or 2 . The case $m = 1$ is impossible: every vertex would lie in exactly one Delsarte clique, and connectedness would force the graph to be complete, contrary to $d \geq 3$. Thus $m = 2$, and Pyber–Skresanov Proposition 2.15 gives $\text{motion}(X) \geq n/16 > \gamma n$. $\square$

7. AN ANALYTIC HAMMING-STABILITY LEMMA

We now handle the only branch not covered by a published endgame with the desired parameter scale.

Proposition 7.1 (Sharpened $\mu = 2$ endgame). *Let X be a Delsarte-geometric distance-regular graph of diameter $d \geq 3$, with smallest eigenvalue $-m$, where $m \leq d$, and $\mu = 2$. Let*

$$0 < \varepsilon = \frac{2}{12d^3 - 1}.$$

Suppose that, for some $2 \leq t \leq d$,

$$(20) \quad b_t \leq \varepsilon k, \quad c_t < \varepsilon k, \quad \theta \geq (1 - \varepsilon)b_1, \quad k > 12d^3.$$

Then X is a Hamming graph.

Proof. The geometric identities of Kivva’s Lemma 2.17 are

$$(21) \quad c_i = \tau_i \psi_{i-1}, \quad b_i = (m - \tau_i) \left(\frac{k}{m} + 1 - \psi_i \right).$$

Kivva’s Lemma 2.18 gives $\tau_2 \geq \psi_1$. Since $\mu = \tau_2 \psi_1 = 2$, we have

$$(22) \quad \tau_2 = 2, \quad \psi_1 = 1, \quad b_1 = \frac{m-1}{m}k, \quad \lambda = \frac{k}{m} - 1.$$

In particular, every neighborhood graph is a disjoint union of m cliques by Kivva’s Lemma 2.20. Since X is geometric and $\mu = 2$, Kivva’s Lemma 3.10 also supplies an induced quadrangle; hence Terwilliger’s inequality is available wherever it is used below.

Since $\varepsilon < 1/m^2$, Kivva’s Lemma 4.2 gives

$$(23) \quad \tau_i < \tau_{i+1} \quad (1 \leq i \leq t-2).$$

The parameters are integral, so $\tau_i \geq i$ for $i \leq t-1$. Put

$$(24) \quad r = m - \tau_{t-1}.$$

Since $\tau_{t-1} \leq m-1$ and $\tau_{t-1} \geq t-1$,

$$(25) \quad 1 \leq r \leq m - t + 1, \quad 2 \leq t \leq m.$$

Let $(u_i)_{i=0}^d$ be the standard sequence for θ . Define the relative drops

$$(26) \quad y_i = \frac{u_{i-1} - u_i}{u_{i-1}}$$

whenever $u_{i-1} > 0$. The standard-sequence recurrence gives the exact identity

$$(27) \quad y_{i+1} = \frac{k - \theta + c_i y_i / (1 - y_i)}{b_i}.$$

Set

$$(28) \quad B = 1 + (m-1)\varepsilon, \quad A = \frac{1 + (5m/3 - 1)\varepsilon}{1 - m\varepsilon}, \quad \delta = A - 1 = \frac{(8m-3)\varepsilon}{3(1-m\varepsilon)}.$$

From (22) and the eigenvalue hypothesis,

$$(29) \quad k - \theta \leq \frac{k}{m}B, \quad u_1 = \frac{\theta}{k} \geq (1-\varepsilon)\frac{m-1}{m} =: u_*.$$

The small-parameter estimates (4) and $m \leq d$ imply

$$(30) \quad m\varepsilon < \frac{1}{50}, \quad A < \frac{6}{5}, \quad \delta < 3d\varepsilon.$$

Indeed, writing $x = m\varepsilon < 1/50$,

$$A \leq \frac{1 + 5x/3}{1-x} < \frac{155}{147} < \frac{6}{5}, \quad \delta < \frac{8d\varepsilon}{3(1-x)} < \frac{400}{147}d\varepsilon < 3d\varepsilon.$$

At $i = 1$, using $c_1 = 1$, $b_1 = (m-1)k/m$, $1/k < \gamma$, and (29),

$$(31) \quad \begin{aligned} y_2 &\leq \frac{B}{m-1} + \frac{m\gamma}{m-1} \frac{1-u_*}{u_*} \\ &= \frac{1}{m-1} \left(B + \frac{m\gamma B}{(m-1)(1-\varepsilon)} \right). \end{aligned}$$

Since

$$\frac{B}{m-1} = \frac{1}{m-1} + \varepsilon \leq 1 + \varepsilon,$$

while $(1+\varepsilon)/(1-\varepsilon) < 2$ and $2\gamma < \varepsilon$, we have

$$\frac{\gamma B}{(m-1)(1-\varepsilon)} < \varepsilon.$$

Furthermore

$$A - B = \frac{m\varepsilon(B + 2/3)}{1 - m\varepsilon} > m\varepsilon.$$

Hence

$$(32) \quad 0 \leq y_2 < \frac{A}{m-1}.$$

If $t = 2$, the product below is empty and only the already positive entry u_1 is used. If $t \geq 3$, then $m \geq 3$, so (32) and $A < 6/5$ give $y_2 < 3/5 < 1$ and hence $u_2 > 0$.

We claim inductively that

$$(33) \quad 0 \leq y_{i+1} \leq \frac{A}{m - \tau_i} \quad (1 \leq i \leq t-2).$$

The base is (32). Suppose $2 \leq i \leq t-2$ and the preceding bound holds. Strict growth gives

$$m - \tau_{i-1} \geq r + t - i \geq 3,$$

so $y_i \leq A/3 < 2/5$ and therefore $y_i/(1-y_i) < 2/3$. Also $c_i \leq c_t < \varepsilon k$ and, since $\psi_i \leq c_{i+1} \leq c_t$,

$$b_i \geq (m - \tau_i) \frac{k}{m} (1 - m\varepsilon).$$

Equation (27) now gives

$$y_{i+1} \leq \frac{B + (2m/3)\varepsilon}{(m - \tau_i)(1 - m\varepsilon)} = \frac{A}{m - \tau_i}.$$

This proves the claim and, in particular, proves that all standard-sequence entries used below are positive.

Using (23), for $2 \leq i \leq t-1$ we have $m - \tau_{i-1} \geq r + t - i$. Therefore

$$\begin{aligned}
 u_{t-1} &= u_1 \prod_{i=2}^{t-1} (1 - y_i) \\
 &\geq u_1 \prod_{s=r+1}^{r+t-2} \left(1 - \frac{A}{s}\right) \\
 &= u_1 \frac{r}{r+t-2} \prod_{s=r+1}^{r+t-2} \left(1 - \frac{\delta}{s-1}\right) \\
 (34) \quad &\geq u_1 \frac{r}{r+t-2} (1 - \delta H_{t-2}),
 \end{aligned}$$

where $H_j = 1 + 1/2 + \dots + 1/j$ and $H_0 = 0$. The final inequality is the elementary product bound $\prod (1 - x_i) \geq 1 - \sum x_i$.

We next show that the t -th sphere contains almost all vertices. Put

$$(35) \quad q = \frac{m\varepsilon}{1 - m\varepsilon} < 1.$$

For $2 \leq i \leq t$, (21) gives

$$\frac{k_{i-1}}{k_i} = \frac{c_i}{b_{i-1}} \leq \frac{q}{m - \tau_{i-1}} \leq q.$$

The remaining first ratio is $k_0/k_1 = 1/k < \gamma < q$; here $q > m\varepsilon \geq 2\varepsilon > 4\gamma$. Thus the entire left tail has mass at most $k_t q / (1 - q)$.

For $t \leq i \leq d-1$, monotonicity gives $b_i \leq b_t \leq \varepsilon k$. Since $m - \tau_i \geq 1$, the second identity in (21) implies

$$\psi_i \geq \frac{k}{m} + 1 - \varepsilon k > k \left(\frac{1}{m} - \varepsilon \right).$$

Consequently

$$\frac{k_{i+1}}{k_i} = \frac{b_i}{c_{i+1}} \leq \frac{\varepsilon k}{\psi_i} \leq q,$$

so the right tail also has mass at most $k_t q / (1 - q)$. Hence

$$(36) \quad \frac{n}{k_t} \leq 1 + \frac{2q}{1 - q} = \frac{1 + q}{1 - q}, \quad k_t \geq (1 - 2m\varepsilon)n.$$

Since $\psi_{t-1} \geq 1$, the second identity in (21) gives $b_{t-1} \leq rk/m$. Combining (29), (34), and (36), we obtain

$$(37) \quad k_{t-1} u_{t-1}^2 \geq \frac{n}{k} FR,$$

where

$$(38) \quad F = (1 - 2m\varepsilon)(1 - \varepsilon)^2 (1 - \delta H_{t-2})^2, \quad R = \frac{c_t r (m-1)^2}{m(r+t-2)^2}.$$

We now prove that, outside one endpoint, R provides a uniform gain:

$$(39) \quad R \geq \frac{m+1}{m}.$$

Kivva's argument gives $c_t \geq t-1$. If $c_t = t-1$, then equality forces

$$\tau_{t-1} = t-1, \quad c_t = c_{t-1}.$$

Kivva's Corollary 2.8 and Terwilliger's inequality then give $4 \leq t \leq m-1$. Thus $r = m - t + 1$ and

$$R = \frac{(t-1)(m-t+1)}{m} \geq \frac{m+1}{m};$$

indeed, the two factors sum to m , the first is at least 3, the second at least 2, and their product is at least $2(m-2) \geq m+1$.

Suppose next that $c_t \geq t$ and $t < m$. For fixed m, t , the function

$$f(r) = \frac{r}{(r+t-2)^2}$$

has no interior minimum on $1 \leq r \leq m-t+1$, so it is enough to check the endpoints. At $r=1$,

$$R \geq \frac{t(m-1)^2}{m(t-1)^2} \geq \frac{m+1}{m},$$

because $t/(t-1)^2$ decreases with t and

$$(m-1)^3 - (m+1)(m-2)^2 = 3m-5 > 0.$$

At $r = m-t+1$,

$$R \geq \frac{t(m-t+1)}{m} \geq \frac{m+1}{m},$$

since the concave function $t(m+1-t)$ has endpoint minimum $2(m-1) \geq m+1$ on $2 \leq t \leq m-1$.

Finally suppose $t = m$, so $r = 1$ and $R = c_t/m$. If $t < d$, the equality $c_t = t = m$ is impossible. Indeed, strict growth gives $c_{t-1} \geq t-1 = m-1$, while $b_t \geq 1$; hence $c_{t-1} - c_t + b_t \geq 0$. Terwilliger's inequality and (21) then give

$$\frac{k}{m} \geq b_{t-1} \geq c_{t-1} - c_t + b_t + \lambda + 2 \geq \frac{k}{m} + 1,$$

a contradiction. Hence $c_t \geq m+1$. If $t = m = d$, then every nonendpoint case also has $c_t \geq m+1$. This proves (39) unless

$$(40) \quad c_t = t = m = d.$$

It remains to show that the loss factor satisfies

$$(41) \quad F > \frac{m}{m+1}.$$

Let $z = \delta H_{t-2}$. We use the elementary bound

$$(42) \quad H_{d-2} \leq \frac{d}{2} \quad (d \geq 3).$$

It holds at $d=3$, and the induction step follows from $H_{d-1} = H_{d-2} + 1/(d-1) \leq d/2 + 1/2 = (d+1)/2$. Thus, by (30),

$$0 \leq z < \frac{3}{2}d^2\varepsilon < 1.$$

Applying $\prod_j (1-x_j) \geq 1 - \sum_j x_j$ to the five factors defining F gives

$$F \geq 1 - (2m\varepsilon + 2\varepsilon + 2z).$$

Since $m \leq d$,

$$2m\varepsilon + 2\varepsilon + 2z < (3d^2 + 2d + 2)\varepsilon < \frac{1}{d+1} \leq \frac{1}{m+1}.$$

The middle strict inequality is equivalent to

$$6d^3 - 10d^2 - 8d - 5 > 0;$$

a completely explicit certificate is obtained by writing $d = x + 3$:

$$6d^3 - 10d^2 - 8d - 5 = 6x^3 + 44x^2 + 94x + 43 > 0.$$

This proves (41) without a finite case split.

Outside the endpoint (40), equations (39) and (41) give $FR > 1$. Hence (37) yields

$$k_{t-1}u_{t-1}^2 > \frac{n}{k}.$$

By Biggs' multiplicity formula, the multiplicity f_1 of θ satisfies $f_1 < k$. Terwilliger's local-eigenvalue theorem, Kivva Theorem 4.1, then puts the eigenvalue

$$-1 - \frac{b_1}{\theta + 1} < -1$$

in every neighborhood graph. This is impossible because each neighborhood graph is a disjoint union of cliques, whose least eigenvalue is -1 . Therefore the endpoint (40) must hold.

At the endpoint, Kivva's endpoint argument (using Lemma 4.2 together with $\tau_d = m = d$) gives $\tau_i = i$ for all i , and $c_d = d = \tau_d \psi_{d-1}$ gives $\psi_{d-1} = 1$. If some ψ_j were at least 2, choose j maximal with that property. Since $\psi_{d-1} = 1$ and every ψ_i is a positive integer, putting $i = j + 1$ gives $\psi_{i-1} \geq 2$ and $\psi_i = 1$ with $2 \leq i \leq d - 1$. Monotonicity of the c_i would give

$$i + 1 = c_{i+1} = \tau_{i+1} \psi_i \geq c_i = \tau_i \psi_{i-1} \geq 2i,$$

which is impossible. Hence $\psi_i = 1$ for every i . Consequently

$$c_i = i, \quad b_i = (d - i) \frac{k}{d},$$

which is the intersection array of $H(d, 1 + k/d)$. Egawa's classification gives a Hamming or Doob graph. A Doob graph occurs only when $1 + k/d = 4$, whereas $k > 12d^3$; hence X is Hamming. $\square$

8. COMPLETION OF THE PROOF

Proof of Theorem 1.1. Suppose that a nonidentity automorphism g has support density $\rho < \gamma$. Proposition 4.1 makes X Delsarte-geometric with $m \leq d$. Lemma 5.1 rules out a transition, so (18) holds; Lemma 5.2 gives (19).

If $\mu \geq 3$, Proposition 6.1 makes X Johnson. If $\mu = 2$, Proposition 7.1 makes X Hamming. If $\mu = 1$, Proposition 6.2 contradicts $\rho < \gamma$. These cases exhaust the positive integer μ . Therefore every graph outside the Johnson and Hamming families has motion at least $\gamma n = n/(12d^3)$. $\square$

9. DEPENDENCY AND AUDIT LEDGER

The proof imports the following published statements.

- (1) From Pyber–Skresanov [1]: the Delsarte clique bound and geometric integrality (Lemmas 2.2–2.3), $2\lambda \leq k + \mu$ (Lemma 2.4), Bang–Koolen geometricity (Proposition 2.5), the full Metsch expression in the proof of Proposition 2.6, the geodesic-load argument and support-sensitive final inequality in Proposition 2.8, Propositions 2.10 and 2.12–2.15, and the Johnson endgames Propositions 2.19–2.20.
- (2) From Kivva [2] (journal numbering): Biggs' formula (Theorem 2.10), the geometric identities and local structure (Lemmas 2.17–2.20), $c_3 > c_2$ (Corollary 2.8), Terwilliger's inequality (Theorem 2.6), the induced-quadrangle lemma (Lemma 3.10), the strict τ_i growth lemma (Lemma 4.2), Terwilliger's local-eigenvalue theorem (Theorem 4.1), and the Hamming/Doob endpoint classification (Theorem 2.25 and the endpoint argument in Theorem 4.7).

Interface	Hypotheses checked in the manuscript
Primitive relation bound	Every nontrivial relation of a primitive rank- $(d+1)$ distance scheme is connected and has diameter at most d .
Metsch expression	$\lambda^2 > 4k\mu$ follows from (17); the ceiling term is not discarded.
Bang–Koolen	The clique bound first gives $m < d+1$, and only then is $m^2\mu < \lambda$ invoked.
Babai spectral motion	The common-neighbor maximum is λ because $\lambda > \mu$.
Johnson endgame	$d > 2$, $\mu \geq 3$, connected neighborhoods, $\varepsilon < 0.0065$, and $k \geq \max\{m^3, 29\}$.
Strict τ growth	$\mu = 2$, $c_t < \varepsilon k$, and $\varepsilon < 1/m^2$.
Terwilliger local eigenvalue	Biggs' formula gives the strict inequality $f_1 < k$, and $\theta > 0$.
Hamming endpoint	The only surviving case is exactly $c_t = t = m = d$; the Doob valency $k = 3d$ is excluded.
$\mu = 1$ motion	Delsarte geometry, Proposition 3.1, $\xi \leq k(1 - 1/d^2)$, and $k > 4md^2, m^2$.

Remark 9.1. The exponent 3 is intrinsic to the present structural route. Small support density ρ gives $\mu \lesssim \rho k$, while the clique argument gives $m = O(d)$ and $\lambda \gtrsim k/d$. The Bang–Koolen condition $m^2\mu < \lambda$ therefore naturally asks for $\rho = O(d^{-3})$. Improving the exponent appears to require a stronger route to geometricity, not merely better constants in the Hamming endgame.

REFERENCES

- [1] L. Pyber and S. V. Skresanov, *On the automorphism group of a distance-regular graph*, J. Combin. Theory Ser. B **172** (2025), 94–114; arXiv:2312.00383.
- [2] B. Kivva, *A characterization of Johnson and Hamming graphs and proof of Babai's conjecture*, J. Combin. Theory Ser. B **151** (2021), 339–374; arXiv:1912.11427.